A CNN Approach for Fine-Grained Dog Breed Recognition

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Abstract

Fine-grained dog breed classification is difficult due to subtle inter-class differences and large intra-class variation. Most prior work reports high overall accuracy on the Stanford Dogs Dataset, but rarely examines the breeds that their models consistently misclassify. This study focusses on improving performance for such weak classes using a ResNet-50 baseline and targeted refinements, including data augmentation, bounding-box cropping, and a feature-level fusion model with Xception. The final ResNet-50 model achieves 85.2% test accuracy, while the fusion model reaches 90.5%. Saliency analysis further shows why certain breeds, such as the Siberian Husky and Canadian Eskimo Dog, remain challenging. The results demonstrate that targeted adjustments can improve classification for visually similar breeds without relying on excessively large or complex architectures.

1 Introduction

Distinguishing between dog breeds is a challenging task due to the large number of breeds and the subtle visual differences between some of them. Images of the same breed can exhibit substantial *intra-class variation*, including differences in age, pose, fur colour or structure, and background distractions. At the same time, certain breeds share strong *inter-class similarity*, with nearly identical physical features that differ only in small local details such as the shape of the ears or fur patterns. This combination of high intra-class variation and subtle inter-class differences makes fine-grained dog breed classification particularly difficult.

Accurate identification of dog breeds has practical applications in veterinary care, breeding programs, and research into breed-specific traits and health conditions. However, prior studies often report strong overall accuracy while failing to analyse performance on breeds that are visually similar or underrepresented. In practice, a model that performs well overall but struggles with these weak or overlapping dog breed classes may be of limited use in real-world application.

This study specifically targets the classification of dog breeds that are commonly misclassified due to subtle inter-class differences or high intra-class variation. To address these challenges, we introduce targeted improvements, including data augmentation, bounding box annotations, and feature-level fusion between a ResNet-50 and Xception architectures. By focusing on breeds that are most frequently misclassified, our proposed method aims to develop a model that generalises effectively across all breeds, instead of further improvements for breeds that are visually distinct or well-represented.

The remainder of this paper is organised as follows: section 2 reviews prior studies and highlights how they handle (or overlook) weak classes, section 3 describes the dataset, experimental setup, and model improvements, section 4 presents the experimental outcomes, and section 5 discusses the implications, limitations of the findings, and potential future direction.

2 Related Work

Several studies have addressed dog breed classification using the Stanford Dogs Dataset[Khosla et al., 2011] or the closely related Tsinghua Dogs Dataset [Zou et al., 2020]. For instance, Cui et al. [2024] developed a model combining multiple CNN architectures, achieving high overall accuracy (95.24%). Importantly, this work also reported inter-class performance, showing that 16 breeds achieved less than 80% accuracy and 6 breeds in the range of 50% and 75%. This demonstrates that models with high aggregate performance can still struggle with specific, visually similar breeds, aligning closely with our focus on improving classification for breeds with high inter-class similarity.

Other studies, such as Valarmathi et al. [2023], compared the performance of various CNN and hybrid architectures on the Stanford Dogs Dataset, achieving high overall validation accuracies. However, these works primarily emphasize aggregate performance and do not explicitly analyse or target breeds that are commonly misclassified due to high inter-class similarity.

Khan et al. [2025] proposed a two-phase pipeline for dog breed recognition, first using YOLO v8N to detect both the dog's head and full body, and then processing these regions with a dual-stream Vision Transformer to extract breed-specific features from each part. The approach achieved strong overall accuracy (90.04%), highlighting the benefit of analysing multiple regions to capture subtle inter-class differences, suggesting that multi-region feature extraction can aid discrimination between visually similar breeds, which relates to our focus on improving classification for weak classes.

In contrast, our proposed approach explicitly targets the classification of dog breeds that are frequently misclassified because of subtle inter-class differences. The objective is to enhance the model's generalisation in real-world applications by achieving higher accuracy for these breeds. We use a ResNet-50 as the baseline model to understand the network's learning behaviour, followed by targeted refinements including data augmentation, bounding box annotations, class balancing, unfreezing top encoder-layers, and feature-level fusion with Xception. By focusing on weak or overlapping classes, our methods aim to create a model that generalises more effectively to breeds that are challenging to classify, addressing a gap largely unexamined in prior related work.

3 Methodology

3.1 Stanford Dogs Dataset

The Stanford Dogs Dataset [Khosla et al., 2011] contains 20,580 images across 120 dog breeds, with each image annotated with a class label. Images within a breed exhibit substantial intra-class variation in terms of age, pose, colour, and background, while some breeds show high inter-class similarity, such as the Papillon and Japanese Spaniel, which have similar fur colours but distinct facial features. This combination of large intra-class variation and subtle inter-class differences makes fine-grained dog breed classification particularly challenging. Additionally, the dataset is imbalanced, as illustrated in Figure 1, which may bias models toward the majority classes.

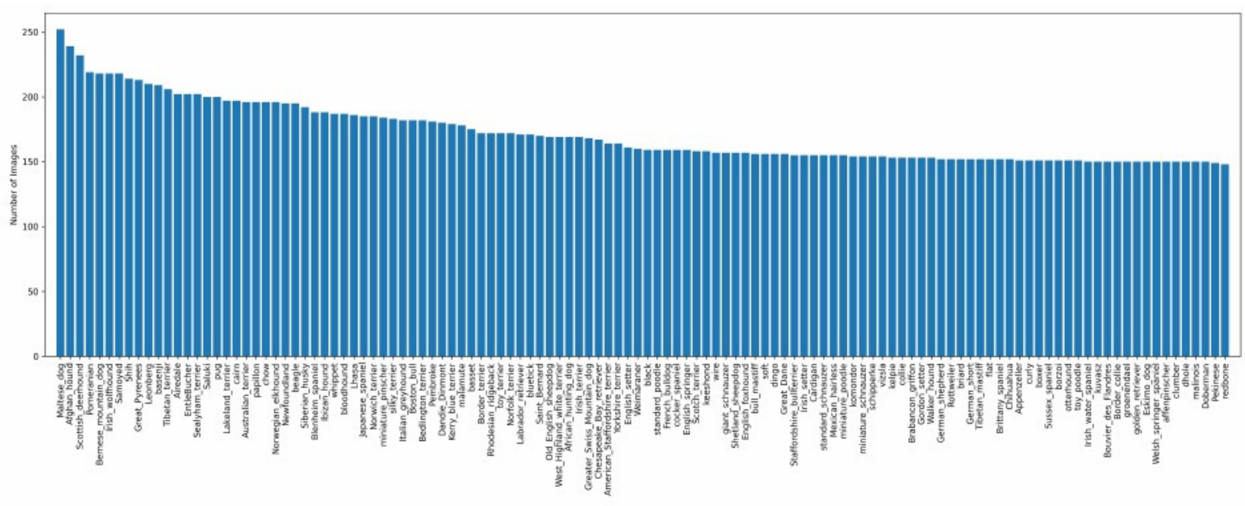

Figure 1: The distribution of images for each dog breed in the Stanford Dogs Dataset.

The Redbone and Pekingese breeds have the fewest images in the dataset, with 148 and 149 samples, respectively, whereas the Maltese and Afghan Hound have 252 and 239 images. This pronounced class imbalance could introduce a bias toward predicting majority breeds, potentially reducing the model's generalisation for minority classes and highlighting the importance of class balancing. For initial experiments, a 25% random subset of each breed was used, resulting in 5,145 images, which were split using a 80/20 split, into 4,116 training samples, 514 validation samples, and 515 test samples. Throughout the experiments, an 80/20 split was used. This smaller subset facilitated faster training during the baseline and early overfitting mitigation experiments (subsection 3.2 and subsubsection 3.3.1). From the Increased Data experiments onward (subsubsection 3.3.2), the full dataset is used.

3.2 Baseline Model

The baseline model employs ResNet-50 [He et al., 2016] as a convolutional encoder, initialised with ImageNet-pretrained weights. It uses residual blocks with skip connections to extract hierarchical features for classification. For the baseline model, this transfer learning setup reuses features learnt on ImageNet for the dog breed classification task. The final fully connected layer of ResNet-50 is replaced with a classification head to predict 120 dog breed classes. The model is compiled with categorical cross-entropy loss, paired with a softmax activation function in the classification head, and optimised using stochastic gradient descent (SGD) with a constant learning rate of 0.03 and a batch size of 32. Architectural modifications to the baseline model are detailed in subsequent experiments.

Training and validation accuracy and loss curves for the baseline model are illustrated in Figure 2. As observed, the model exhibits substantial overfitting: for the last epoch, the training accuracy reaches 0.9990 while the validation accuracy is 0.5098, and the training loss is 0.0697 compared to a validation loss of 1.8168. The validation loss reaches its minimum at epoch 10 and then subsequently increases, indicating that the model begins to memorise the training data rather than generalise to validation samples. To address potential optimisation issues, a ReduceLROnPlateau learning rate scheduler was applied, reducing the learning rate by half if no improvement in the validation loss was observed over three epochs, with a lower bound of $1 \cdot 10^{-6}$. This adjustment helps stabilise the training process by allowing finer convergence toward a local minimum around 1.7560, but has no effect on the severe overfitting.

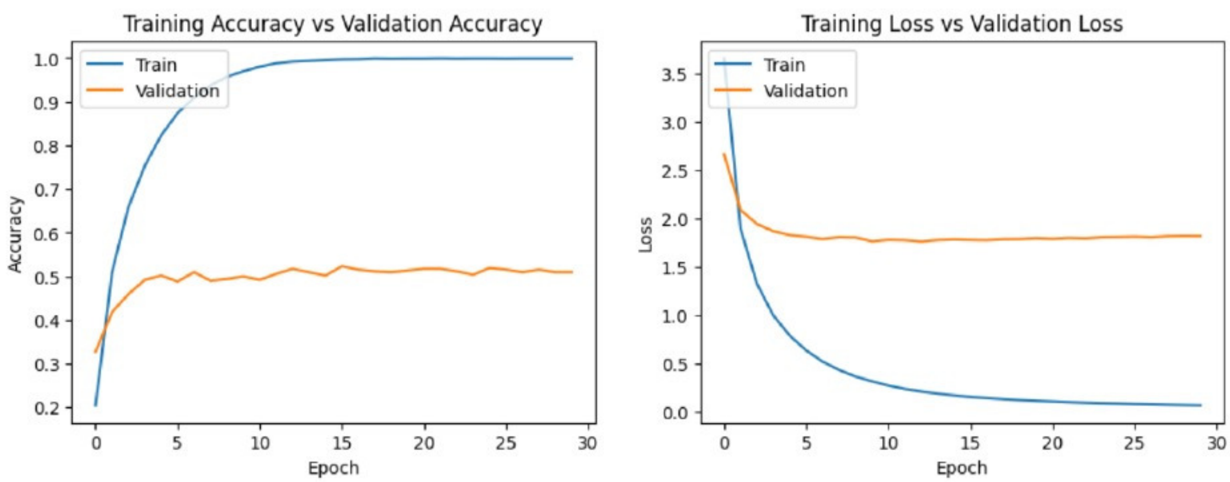

Figure 2: Loss and accuracy curves for the baseline model.

3.3 Experiments

3.3.1 Mitigating Overfitting

The baseline model, consisting of a single dense classification head atop the frozen ResNet-50 encoder, exhibited substantial overfitting. To address this, we first introduced additional dense layers with kernel regularization, systematically evaluating L1 (lasso), L2 (ridge), and elastic net (EN) penalties. This approach, motivated by prior work [Farhadi et al., 2022], aims to constrain weight magnitudes and reduce co-adaptation among neurons, thereby improving the robustness of the learned features. Subsequently, dropout was incorporated as an alternative regularization technique, also motivated by Farhadi et al. [2022], which randomly deactivates a subset of neurons during training to encourage redundant and more generalisable feature representations. Initial experiments with a single dropout layer with different rates, a dropout rate of 0.6 achieved training and validation accuracies of 0.9194 and 0.5725, respectively; increasing the dropout rate beyond that reduced validation performance. To mitigate this, dropout was distributed across two hidden layers with lower rates (0.5 and 0.4), reducing the gap between training and validation accuracies to 0.7929 and 0.5490, respectively. The additional layer might potentially also increase the models representational capacity. The network architecture for this model, reflecting the added layers in between pooling and the final fully connected layer is shown in Table 1.

Table 1: Modified ResNet-50 Baseline Model Architecture

Component	Layer Type	Output Shape	Parameters	Trainable
Encoder	ResNet-50	(7, 7, 2048)	23,587,712	No
Decoder	GlobalAveragePooling2D	(2048,)	0	Yes
	Dense1 (ReLU)	(512,)	1,049,088	Yes
	Dropout1 (0.5)	(512,)	0	Yes
	Dense2 (ReLU)	(256,)	131,328	Yes
	Dropout2 (0.4)	(256,)	0	Yes
	Fully-connected (Softmax)	(120,)	30,840	Yes

3.3.2 Increased Data Diversity

Following the observed increase in validation accuracy achieved by randomly disabling neurons via dropout, we implemented data augmentations on the subset of the dataset (25% of each dog breed) to further encourage the model to learn by introducing additional variability and increasing data diversity, thus improving generalisation. This resulted in substantial improvements and a further reduction in overfitting, with a training accuracy of 0.8123 and a validation accuracy of 0.7588, and training and validation losses of 0.6992 and 0.7440, respectively.

Motivated by the substantial improvements achieved with data augmentation on the 25% subset of the dataset, we hypothesised that applying data augmentations to the full dataset could further enhance model performance. All subsequent experiments are using the full dataset. Additionally, the deeper decoder architecture with two hidden layers, introduced in the previous experiment, has contributed to the improved results by providing greater capacity to learn from the increased training data. Training the model on the full dataset with these augmentations resulted in a training accuracy of 0.7827 and a validation accuracy of 0.797, and a training and validation loss of 0.5992 and 0.6717, respectively. Indicating both improved generalisation and the absence of overfitting. To verify how the model generalises, we evaluate the model performance on the unseen test data, resulting in a accuracy of 0.7901.

To identify overlapping dog breed misclassifications, we use the F1-score, which combines precision and recall via their harmonic mean. Dog breed classes with an F1-score below 0.70 are considered as weak classes. The results reveal that the model struggles with 28 breeds in total. To further examine classification difficulties arising from inter-class similarity, we inspect the confusion matrix illustrated in Figure 3, which demonstrates that the model encounters substantial misclassifications among visually very similar breeds; there are 14 misclassifications with 2 or more instances of overlaps for similar breeds. Notable pairs or groups include: Eskimo Dog $\leftrightarrow$ Siberian Husky $\leftrightarrow$ Alaskan Malamute, Toy Poodle $\leftrightarrow$ Miniature Poodle $\rightarrow$ Standard Poodle, Walker Hound $\leftrightarrow$ English Foxhound, Collie $\leftrightarrow$ Shetland Sheepdog, Whippet $\leftrightarrow$ Italian Greyhound, and Standard Schnauzer $\leftrightarrow$ Miniature Schnauzer.

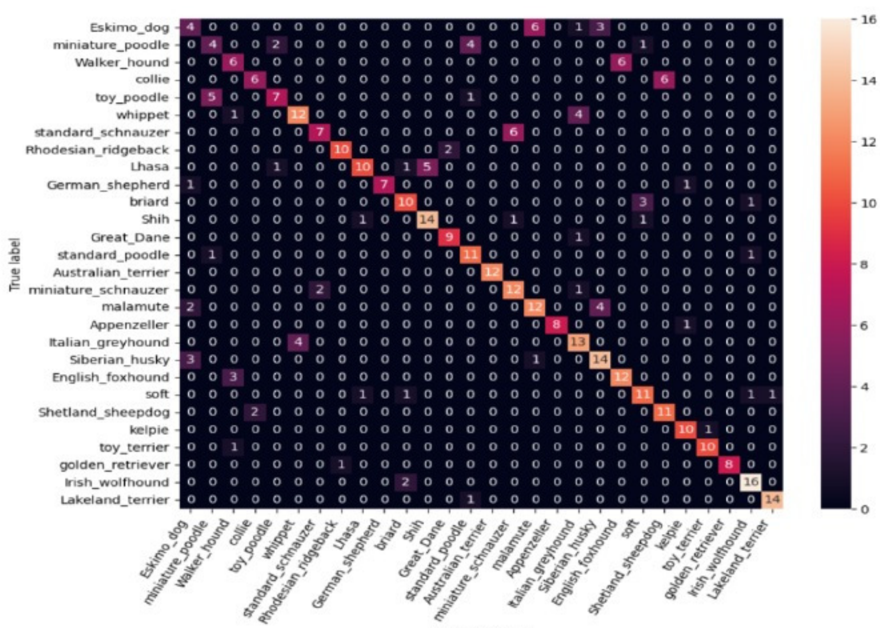

Figure 3: Confusion matrix for ResNet-50 model (Exp. 4) for dog breeds with an F1-score ≤ 0.70

3.3.3 Addressing Inter-class Misclassifications

To better understand the causes of inter-class misclassifications, we inspected misclassified test images for the affected dog breeds and also manually inspected their respective image directories. Misclassifications appear to arise from distractions in the images, such as multiple dogs, toys resembling dogs, or humans. Additional challenges occur with visual similarities between distinct breeds, including shared hairstyles and textures, as seen in Miniature Poodles and Toy Poodles. Intra-breed variations, such as the presence of both puppies and adult dogs in the dataset, may also contribute. Examples of some of the potential causes identified are shown in Figure 4.

Figure 4: Images that show potential causes for inter-class misclassifications.

To reduce inter-class misclassifications, the class imbalance inherent in the dataset was addressed by applying class weights during training, computed directly from the distribution of training labels to increase the gradient contribution of under-represented breeds. Although this adjustment did not improve the overall model performance, improvements were observed for the identified weak dog breed classes.

Expanding on the inter-class misclassifications, prior work mentions bounding-box annotations provided with the dataset [Khosla et al., 2011], which are likely to mitigate several of the identified misclassification causes: humans, dog-like toys, multiple dogs, and other distracting elements. Using the bounding box annotations to crop the images resulted in substantial performance improvements, yielding a training accuracy of 0.8477 and a validation accuracy of 0.8474, with corresponding training and validation losses of 0.4663 and 0.5078. Evaluation on the test data confirmed these gains, achieving a test accuracy of 0.8523, a test loss of 0.4848, a recall of 0.8492, and an F1-score of 0.8483. More importantly, the number of inter-class misclassifications, defined by an F1-score below 0.70, was reduced to 13. The no. of instances overlapping misclassifications with 2 or more occurrences is decreased to 7 now, observable in the confusion matrix in Figure 5.

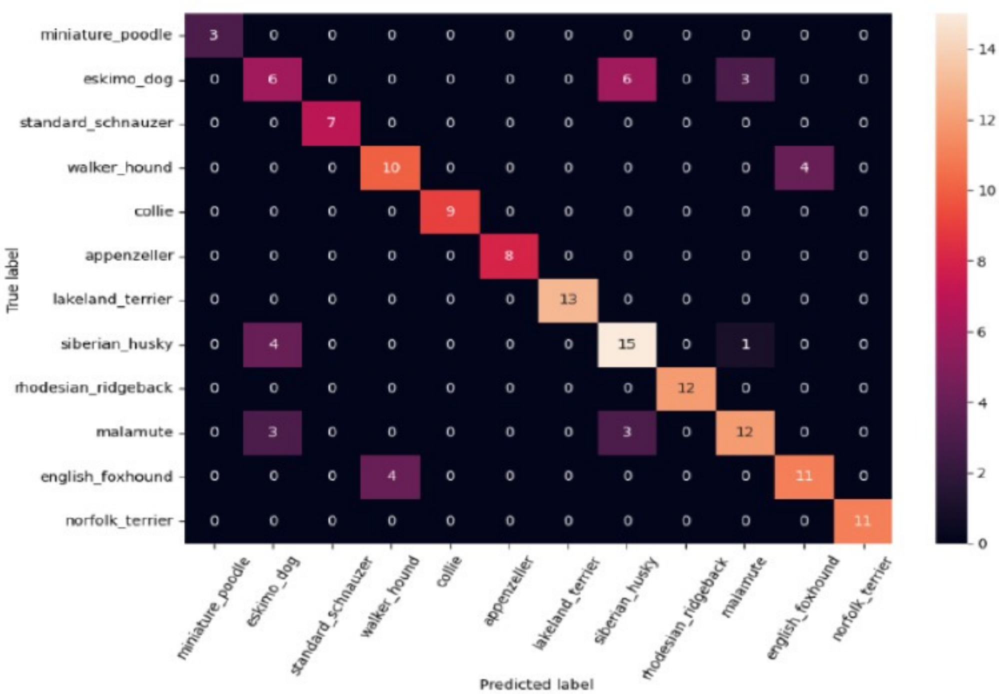

Figure 5: Confusion matrix for ResNet-50 model (Exp. 6) for dog breeds with an F1-score ≤ 0.70

3.3.4 Fusion Model

Experiments with unfreezing the top layers of the encoder, did not result in improvements. Inspired by Valarmathi et al. [2023], who demonstrated that fusion models enhance fine-grained dog breed recognition, we developed a fusion model combining our ResNet-50 model with a Xception model. Xception was selected due to its architectural differences from ResNet-50, providing complementary feature representations [Valarmathi et al., 2023]. For the Xception decoder, batch normalization with ReLU activation was applied between dense layers and dropout, motivated by the encoder's use of separable convolutions followed by batch normalization and ReLU activation [Chollet, 2017]. Prior to implementing a feature-level fusion, a soft-voting mechanism using the predictions from Xception for the identified weak classes was used to validate the potential benefits. The model reduced total weak class misclassifications from 83 to 57.

Based on these results, a feature-level fusion model was implemented. The output features from both ResNet-50 and Xception branches were concatenated and passed through batch normalization and ReLU activation to align feature scales. To ensure proper model training, each branch received input images normalised according to the range expected by its pretrained encoder (ResNet-50: [0, 1] and Xception: [-1, 1]), and the corresponding data generators were synchronised to provide matching batches with shared labels. A shared dense layer preceded the softmax classifier, allowing the model to learn a joint representation of combined features. The architecture is summarized in Table 2.

Table 2: Fusion Model (ResNet-50 + Xception) Layer Summary

Component	Layer Type	Output Shape	Parameters	Trainable
Encoder 1	ResNet-50	(7, 7, 2048)	23,587,712	No
Encoder 2	Xception	(7, 7, 2048)	20,873,856	No
Decoder 1	GlobalAveragePooling2D	(2048,)	0	Yes
	Dense1 (ReLU)	(512,)	1,049,088	Yes
	Dropout1 (0.5)	(512,)	0	Yes
	Dense2 (ReLU)	(256,)	131,328	Yes
	Dropout2 (0.4)	(256,)	0	Yes
	Fully-connected (Softmax)	(120,)	30,840	Yes
Decoder 2	GlobalAveragePooling2D2	(2048,)	0	Yes
	Dense3 (linear)	(1024,)	2,097,152	Yes
	BatchNormalization1 (ReLU)	(1024,)	4,096	Yes
	Dropout3 (0.45)	(1024,)	0	Yes
	Dense4 (linear)	(512,)	524,800	Yes
	BatchNormalization2 (ReLU)	(512,)	2,048	Yes
	Dropout4 (0.40)	(512,)	0	Yes
	Concatenate	(768,)	0	Yes
Fusion	BatchNormalization3 (ReLU)	(768,)	1,536	Yes
	Dense5 (ReLU)	(256,)	196,864	Yes
	Fully-connected (Softmax)	(120,)	30,840	Yes

Evaluation of the feature-level fusion model yielded a test accuracy of 0.9052, test loss of 0.3070, a recall of 0.9033, and F1-score of 0.9023, indicating improved overall performance. Inter-class misclassifications were further reduced: only four breeds exhibited F1-scores below 0.70: Eskimo Dog (0.400), Siberian Husky (0.651), Miniature Poodle (0.667), and American Staffordshire (0.688), with an average F1-score of 0.601, improved from 0.498 in the previous experiment. The number of overlapping misclassification instances decreased to only 2 breeds, as illustrated in Figure 6

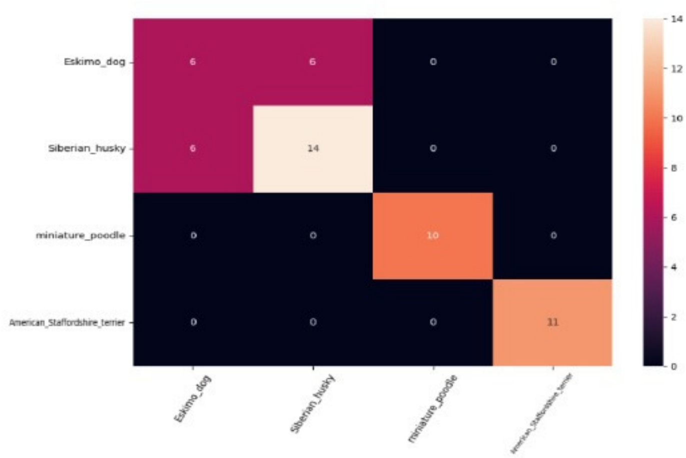

Figure 6: Confusion matrix for fusion model (Exp. 8) for dog breeds with an F1-score ≤ 0.70

4 Results

4.1 Evaluation Metrics

To measure the performance of the model, the following metrics are used: accuracy, precision, recall, and F1-score. The metrics are given as follows:

$$\begin{aligned} \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, & \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}}, & \text{F1-score} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \tag{1}$$

Where TP are true positives, TN are true negatives, FP are false positives, and FN are false negatives. Accuracy is the number of correct predictions the model produces from all the predictions. Precision shows how accurate the true predictions are, while recall is a measure of how well the model detects true positives successfully. The F1-score is the harmonic mean of precision and recall, penalising poorly balanced precision and recall [Khan et al., 2025].

4.2 Model Evaluation

The baseline model achieved a test accuracy of 0.5314, with a precision of 0.528, a recall of 0.532, and an F1-score of 0.530. Subsequent experiments aimed at mitigating overfitting resulted in a modest increase in accuracy to 0.5314, accompanied by a precision of 0.528, a recall of 0.532, and an F1-score of 0.530. Expanding the training data and applying data augmentation substantially improved model performance, yielding a test accuracy of 0.7901, a precision of 0.755, a recall of 0.748, and an F1-score of 0.746, demonstrating markedly enhanced generalisation on unseen data. Further refinements targeting inter-class misclassifications, including data augmentations, addressing class imbalance and leveraging bounding box annotations to reduce distractions in the images, increased the test accuracy to 0.8484, with a precision of 0.845, a recall of 0.851, and an F1-score of 0.848. The final experiments, which incorporated unfreezing of the encoder's top layers and a feature-level fusion approach, achieved the highest performance, with a test accuracy of 0.9052, a precision of 0.903, a recall of 0.907, and an F1-score of 0.9052. These results indicate a significant reduction in misclassifications for the very similar dog breeds, an area where prior models struggled, and highlight the effectiveness of the proposed model in improving generalisation and robustness across the weak classes. Summarised in Table 3.

Table 3: Performance Metrics Across Experiments

Experiment	Test Accuracy	Precision	Recall	F1-score
ResNet-50: Baseline Model	0.5314	0.528	0.532	0.530
ResNet-50 (Exp. 4): Increase Data Diversity	0.7901	0.755	0.748	0.746
ResNet-50 (Exp. 6): Inter-class Improvements	0.8484	0.845	0.851	0.848
Fusion model (Exp. 8): ResNet-50+Xception	0.9052	0.903	0.907	0.9052

4.3 Inter-class Misclassifications

Table 4 presents the improvements in total no. of misclassifications, the number of dog breed classes with an F1-score below 0.70 (defined as the weak classes), and the misclassification counts for these. The results demonstrate a significant improvement in relation to related prior work Cui et al. [2024].

Table 4: Experiment Misclassification Summary

Experiment	Total Misclassifications	No. of Weak Classes ($\leq$ 0.70 F1)	Weak Class Misclassifications
ResNet-50 (Exp. 4): Increase Data Diversity	411	28	130
ResNet-50 (Exp. 6): Inter-class Improvements	321	13	83
Fusion model (Exp. 8): ResNet-50+Xception	195	4	26

4.4 Comparison against Benchmarks

The performance of our proposed ResNet-50-based model and the feature-level fusion model combining ResNet-50 with Xception was compared against the model results from prior work, as summarized in Table 5. The test accuracy of our model does not achieve the same high overall test accuracy as the current state-of-art models.

Table 5: Comparison of Model Performance on the Stanford Dog Breed Dataset

Study	Model/Method	Test Accuracy
[Liu et al., 2016]	ResNet-50	88.9%
[Kamdar et al., 2021]	ResNet-50	90.1%
Our Model	ResNet-50	85.2%
[Valarmathi et al., 2023]	InceptionV3 + Xception	92.4%
[Cui et al., 2024]	InceptionV3 + InceptionResnetV2 + NASNet + PNASNet	95.2%
Our Model	ResNet-50 + Xception	90.5%

5 Discussion

The proposed method successfully reduced inter-class misclassifications compared to related prior work discussed in this report, with data augmentation and feature-level fusion contributing most significantly. In contrast, unfreezing the top layers of the encoder, to allow additional fine-tuning, had minimal impact on the ResNet-50 model and, in fact, negatively affected the performance of the fusion model, where it slightly worsened the inter-class misclassifications.

For experiments using the full dataset with augmentations, the volume of data resulted in very long training times. While this was necessary for the focus of this report on improving inter-class misclassifications of weak classes, it meant that training could not be completed in one session. This led to the data being loaded and split on a couple of occasions. Although the same seed was used, it would have been a better choice to split only once and store these in a directory. Although no changes in performance were observed from rerunning the code, it remains a key point in future iterations to guarantee fairer comparison.

Despite the inter-class improvements, the Eskimo Dog and Siberian Husky remain particularly challenging due to their extreme visual similarity. While the Siberian Husky (F1-score of 0.6512) has support of 192 images, the Eskimo Dog only has 150 images, which might provide some explanation for the issues, but not a justification of an F1-score of only 0.4000. Manually inspecting the dataset for these breeds, reveal that the two breeds are very hard to distinguish. To investigate beyond the high visual similarity between the two breeds and to identify where the model focusses for these. To further investigate the model's behaviour, saliency maps were generated (Figure 7). For correctly classified Eskimo Dogs, the model primarily focused on the face. In contrast, Eskimo Dogs classified as Siberian Huskies, often exhibited attention to the fur pattern around the neck and background elements such as snow, which are more commonly associated with Siberian Huskies in the training data, biasing the predictions. Similarly, correctly classified Siberian Huskies focused on the face and neck pattern, along with background snow cues. Siberian Huskies, misclassified as Eskimo Dogs, often showed increased focus on the face and reduced attention to the neck region and minimal focus on the background. Across both breeds, correct classifications relied on discriminative facial and neck pattern features, while misclassifications were influenced by missing neck patterns or focusing on contextual elements, such as snow in the background.

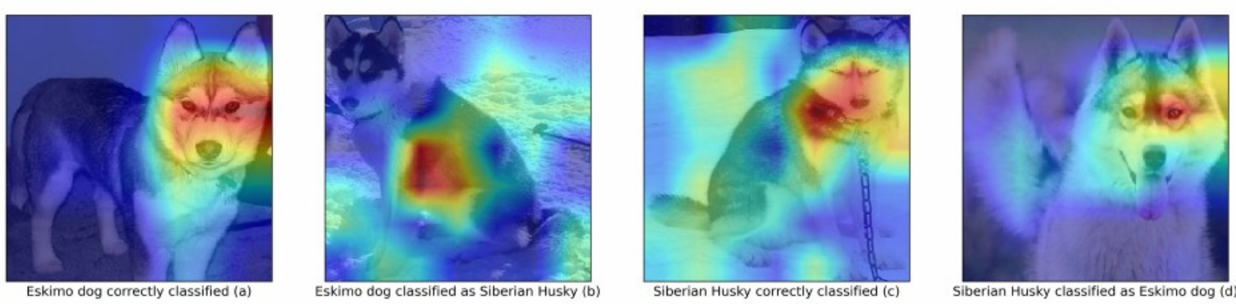

Figure 7: Saliency maps of correct and incorrect classifications for the ResNet-50-based model.

From the saliency maps, it's noticeable that both the final models focus a lot on the faces and areas around the neck, almost entirely neglecting the body. This might suggest that future directions of this study, could include Multi-task Learning (MTL) approach as demonstrated by Khan et al. [2025], to train a model separately on the faces of the dogs and another model on the bodies. The initial step towards this approach was taken, using the Dog Face Detector [Kairess, 2018] to identify the dog's face regions and store these in a separate directory.

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